Chitraksh Sinha

ce23btech11017@iith.ac.in | chitraksh.sinha@gmail.com | +91 9167427509 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.93
Pace Junior College	Class XII (SSC Board)	2023	81.00%
Seth Juggilal Poddar Academy	Class X (ICSE)	2021	98.20%

EXPERIENCE

Traffic Behavior Clustering Using Deep Learning and K-Means | Under Dr. Digvijay Pawar

Technologies - Python, OpenCV, BoTSORT, K-Means, YOLOv8, EVT

- Processed **7h** UAV footage and **15K+** vehicles using **YOLOv8** and **BoTSORT** for trajectory extraction across **3** work zones.
- Developed dual-scale **K-Means clustering** to classify **2** vehicle behaviours and **6** microscopic traffic states from trajectory features.
- Integrated **ACT-POT-EVT** and traffic-state modelling to quantify conflict risk and identify pre-critical traffic sequences.

PROJECTS

Multi-Agent AI News Brief Generator | *FastMCP, Groq, A2A, Streamlit, Rest APIs*

- Engineered a **multi-agent news brief generator** aggregating real-time weather, topic-specific news, and finance data using a modular architecture built on **MCP** and **A2A** agent communication.
- Built **custom MCP servers** plus **specialised scout** and **publisher agents** with FastMCP, handling API integration, schema validation, and coordinated task execution across agents.
- Delivered an end-to-end reporting pipeline using Groq for context-aware article generation and Streamlit for interactive report creation, viewing, and export.

Multi-Agent Research Assistant | *Python, AutoGen, Groq LLM, arXiv API*

- Architected a **multi-agent research assistant** using AutoGen and a Groq-hosted LLM, orchestrating **five specialized agents** to automate literature review and report generation via the **arXiv** API.
- Implemented selector-based multi-agent orchestration with retry mechanisms and **human-in-the-loop** approval checkpoints to improve task-routing reliability.
- Converted natural language research queries into structured literature reviews by integrating LLM reasoning with live academic search results.

Medical Chatbot using RAG | *Python, LangChain, Pinecone, Groq, Hugging Face, Flask, HTML, CSS*

- Built a **domain-specific medical chatbot** using RAG with **LangChain** and **Groq Llama 3.3 70B**, grounding responses in a **4,505**-page medical knowledge base.
- Designed a semantic search pipeline with Hugging Face embeddings and Pinecone, chunking the corpus into 45,050 segments and retrieving the top-**3** most relevant chunks per query.
- Built the Flask backend and HTML/CSS frontend, integrating retrieved context with the LLM and managing credentials via environment variables.

Image Caption Generator | *Python, TensorFlow, Keras, NumPy, VGG16, LSTM, NLP*

- Developed an image caption generation model using **VGG16** and an **LSTM decoder** on the **Flickr8k** dataset (8,092 images).
- Implemented an end-to-end TensorFlow pipeline for **image preprocessing**, **caption tokenization**, **sequence generation,** and model training.
- Achieved **BLEU-1: 0.546** and **BLEU-2: 0.319** for caption generation using autoregressive next-word prediction.

SKILLS

- **Programming & Tools** - Python, SQL, Git, GitHub, VS Code, Jupyter Notebook, Google Colab
- **AI/ML** - Supervised & Unsupervised Learning, Deep Learning, NLP, CNN, RNN/LSTMs, Transformers, RAG, LLMs, Multi-Agent Systems, MCP
- **Frameworks/Libraries** - NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, OpenCV, LangChain, FastMCP, AutoGen
- **CS Fundamentals** - Data Structures & Algorithms, OOPs, DBMS, System Design

RELEVANT COURSES

- **Technical Courses -** Introduction to Programming, Machine Learning Specialization by Andrew Ng, Deep Learning Specialization by Andrew Ng, Complete Generative AI and RAG Course (Krish Naik), Prompt Engineering (Great Learning)
- **Mathematics -** Calculus, Differential Equations, Linear Algebra, Transform Techniques, Probability and Statistics

SCHOLASTIC ACHIEVEMENTS

- Solved over **200** problems on LeetCode, covering topics like Dynamic Programming, Trees, Graphs, Arrays, etc.
- Secured **1st** position in the State Level Creative Writing Contest 2021.
- Secured **98.2%** in the ICSE 10th board exam and was the topper of my school.

EXTRACURRICULAR

Passionate about playing **intermediate chess**, solving **Rubik's Cubes** and logical puzzles like **Sudoku**. Passionate **self-taught guitarist**, drummer and pianist. Secured gold medal in the Battle of Bands in MILAN, the Inter-Hostel sports and cultural event. Passionate about football, cricket, badminton, F1, Clash Royale etc. Actively engage in following each, along with **tactics** and **team dynamics**.